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Discourse-Level Watermarking using Rhetorical Structure Theory

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***Abstract*—Large language model (LLM) text is increasingly hard to distinguish from human writing, but many existing text watermarking schemes remain brittle. In particular, signals encoded in lexical choices can be attenuated or removed by paraphrasing and other meaning-preserving rewrites. To address this, we are first to propose a discourse-level watermarking framework grounded in Rhetorical Structure Theory (RST), where rhetorical relations serve as a stable carrier for attribution under semantic-preserving transformations. This shift strengthens originality by moving beyond surface-form heuristics to discourse organisation, and it strengthens significance by targeting the most common real-world evasion route, namely routine and adversarial paraphrasing encountered in provenance and accountability settings.**

We realise the approach as a prompt-based, black-box pipeline that requires no access to model internals and can be applied to off-the-shelf LLMs. Using the standard in the prior literature metrics, we demonstrate rigour by evaluating detectability at a controlled false positive rate of 1% and assessing robustness under paraphrase attacks across multiple datasets and generation settings, with direct comparison to a strong lexical watermarking baseline (PostMark). Results show competitive pre-attack detection (95 to 99% TPR at 1% FPR) while improving paraphrase robustness by roughly 15 percentage points overall and roughly 18 percentage points on OpenGen, without materially degrading output quality.

***Index Terms*—large language model, watermarking, generative AI, discourse**

I. Introduction

Large language models (LLMs) now generate coherent, contextually rich, and stylistically adaptive text across domains including journalism, education, and scientific communication [1]–[3]. As these systems increasingly emulate human linguistic and rhetorical behaviour, the distinction between human-authored and machine-generated text becomes progressively harder to establish [4]–[6]. This shift has practical and societal consequences. LLM outputs can accelerate misinformation, complicate academic integrity and authorship verification, and introduce risks when future models are trained on synthetic text [5]. Addressing these risks requires technical mechanisms that support verifiable provenance while preserving meaning and quality [7], [8].

Watermarking is a prominent approach to provenance auditing for LLM text. Statistical or lexical watermarking methods embed a hidden signal by biasing token choices, enabling post hoc detection while aiming to preserve fluency [4], [9]–[12]. However, robustness remains a core limitation. Paraphrasing, back-translation, and adversarial rewriting can disrupt token-level regularities while keeping semantics largely intact, thereby reducing watermark detectability [6], [7], [9]. Recent results have sharpened this concern by showing that strong robustness can be difficult to guarantee under realistic editing operations [6]. These challenges motivate exploring watermark carriers beyond surface form.

In this paper, we propose a discourse-level watermarking method grounded in Rhetorical Structure Theory (RST) [13]. Rather than encoding a signal in token distributions, we encode it through constraints on rhetorical relations between discourse units. The underlying premise is that high-quality paraphrases tend to preserve a text's communicative intent and discourse organisation even when they substantially alter wording. This shift in carrier is central to the **originality** of our approach: it leverages discourse structure as a stable channel for attribution under meaning-preserving transformation.

We implement the method as a prompt-based, black-box pipeline that requires no access to model internals and can be applied to off-the-shelf LLMs, enabling practical deployment across model providers. We evaluate detectability using the standard in the prior work metrics: true positive rate (TPR) at a fixed false positive rate (FPR) of 1%. We also quantify the impact of paraphrasing on detection performance. Experiments use GPT-4.1 to generate and score watermarks on three datasets adopted from prior work [14]: OpenGen, LFQA, and FactScore.

Our results indicate that discourse-level watermarking achieves competitive detectability at 1% FPR while improving robustness to paraphrasing relative to a strong prompt-based baseline. In particular, we observe similar pre-attack detection rates and consistently higher post-paraphrase detectability on challenging settings, notably OpenGen, supporting the **significance** of discourse-level carriers in scenarios where rewriting is expected.

In summary, we make three contributions:

- We introduce a discourse-level, RST-based watermarking framework that embeds and verifies provenance signals

through rhetorical relations rather than token distributions.
- We present a practical prompt-based implementation that operates in a black-box setting and is compatible with off-the-shelf LLMs.
- We provide quantitative evaluation and ablation evidence showing competitive detectability and improved robustness to paraphrasing relative to a strong prompt-based baseline.

II. Related Work

A. Watermarking

A good watermark should balance multiple qualities that ensure it is effective, resilient, and unobtrusive [6]–[8], [11], [12], [15]–[20]. While these properties are well established in classical digital watermarking, they take on new significance for text generated by large language models (LLMs), where linguistic and semantic constraints limit the available embedding space:

Effectiveness refers to the ability of a watermark to be reliably detected by authorized systems using its designated verification method. It must consistently identify marked text across topics, styles, and text lengths without false positives or negatives.

Robustness measures the watermark's capacity to withstand intentional or unintentional alterations such as paraphrasing, summarization, translation, or adversarial rewording. A robust watermark remains detectable when the text's surface form changes but its semantic structure is preserved.

Fidelity ensures that watermark embedding does not degrade the text's readability or meaning. High-fidelity watermarks maintain grammatical correctness, natural style, and semantic equivalence between the watermarked and non-watermarked text.

Efficiency concerns the computational cost of both embedding and detection. Practical watermarking approaches should be lightweight, scalable, and suitable for integration with real-time generation workflows.

Undetectability describes the watermark's invisibility to unauthorized users and models. A watermark should not be perceptible by human readers or detectable without access to the secret key or detection protocol, ensuring that verification remains a secure and controlled process [6], [21].

B. Paraphrasing Impact on Watermarking

Paraphrasing is the act of altering a piece of text in a way that retains the original meaning. This is achieved via changing the words and sentences with those of similar meaning, such as changing the word "car" to the word "vehicle". Paraphrasing attacks are when someone tried to alter the text in some way to remove an embedded watermark, and are difficult to defend against. Many current systems use adversarial training to better defend their model against these types of attacks by training them on embedded text that has undergone paraphrasing [22].

C. The Approaches to Text Watermarking

Watermarking text generated by LLM has its many challenges: we can't know how long a generated text structure will be, we don't know what topic it may be on, and we don't know what prior information the user might have provided to the LLM. Compared to watermarking images there is much less usable space (redundancy) to work with, so Cryptography is not a viable option. Instead, there are three main forms of LLM watermarking: 1) rule-based 2) statistical and 3) prompt-based [6].

Rule-based watermarking focuses on altering the text at the word or sentence level. It replaces the words in the sentence with their synonyms from a dictionary. Although it can also alter the syntactic structures of sentences, this approach is obviously vulnerable to paraphrasing and similar rule-based text altering attacks [6].

Statistical (AKA "lexical" or "probabilistic") watermarking represents one of the earliest and most practical approaches to identifying AI-generated text. These methods, exemplified by the work of [4] and subsequent refinements [9]–[12], embed imperceptible statistical signals into the token distribution during generation. By biasing word selection, lexical watermarking enables reliable post hoc detection of generated text while preserving fluency and coherence. Although effective against naive attacks such as random token substitutions, lexical reordering, and simple synonym-based paraphrasing, these techniques face significant challenges when subjected to more sophisticated perturbations including neural paraphrasing, back-translation, or adversarial rewording [6], [7], [9]

Prompt-based watermarking: recent work has proposed a complementary class of prompt-based watermarking approaches [14], [23], in which watermark patterns are embedded through instructional prompts rather than direct manipulation of the model's token probabilities. These methods operate entirely in a black-box setting and rely on the model's own linguistic regularities to carry the watermark signal, enabling watermarking without model access or retraining.

In our study, we adopt a prompt-based watermarking strategy because it enables direct control over the generation process at the discourse level. By structuring prompts to guide rhetorical and relational coherence, the model can embed watermark cues that align with the text's semantic organization rather than its surface form. This approach provides several advantages: it enhances robustness to paraphrasing and reformatting, since the watermark is tied to discourse relations rather than token sequences; it maintains textual fluency and naturalness, as the model operates within its normal generation dynamics; and it allows for *interpretable* watermark patterns, where specific discourse structures can serve as verifiable markers of provenance. For these reasons, we evaluate our discourse-level prompt-based method against other existing prompt-based approaches to assess its relative resilience and interpretability.

D. Rhetorical Structure Theory

Rhetorical Structure Theory (RST) characterises a text in terms of the functional relations that hold between spans of discourse [13]. Table I provides five concise examples of common relations. RST represents a document as linked discourse units, where each link specifies how one span supports, motivates, contrasts with, or otherwise relates to another, and thus captures structure beyond a flat sentence sequence.

A central distinction in RST is between the Nucleus (N) and the Satellite (S). The Nucleus conveys the main claim or action that can stand on its own, while the Satellite provides supporting material such as explanation, justification, or context. The Satellite is typically dependent on the Nucleus for its interpretation [13].

This discourse-level view is useful in our setting because meaning-preserving edits, such as paraphrase, can substantially change surface wording while retaining communicative intent and discourse organisation.

III. Methodology

We conduct experiments to test whether RST can support an effective, non-evasive watermark for AI-generated text. Our objective is a watermarking approach that (1) preserves the quality and meaning of the original text and (2) remains robust under paraphrasing. We use a prompt-based approach that embeds and detects rhetorical structure relations in text.

A high-level overview of the workflow is shown in Figure 1. To improve robustness, we use a revision loop with a maximum of $N = 6$ attempts. At each attempt, a watermark is inserted and then detected (extracted) by comparing the reported score with a specified threshold. We use a prompt-based *inserter* that asks a general-purpose LLM to generate or revise a passage whose discourse structure exhibits a controlled density of selected RST relations. The inserter prompt contains:

- **Relation inventory:** a set of rhetorical relations with brief definitions and associated weights.
- **Structural rules:** the commands that insert several high-weight relations in nucleus/root positions, supported by lower-weight satellite relations in a clear hierarchy.
- **Target strength:** an instruction to meet a watermark-strength objective computed as a length-normalised relation-weight sum (per fixed word budget), reducing spurious inflation for longer passages.

Key prompt parameters (for example, relation weights, the operational definition of *high-value*, and the decision bounds used later for classification) were manually tuned after preliminary experimentation. Full prompt templates and parameter settings are documented in our GitHub repository[1]. If an output is scored as insufficiently watermarked by the downstream scorer and decider, the inserter is re-applied with brief feedback (for example, increasing nucleus high-value relations or adjusting relation diversity) until the target strength is met or the attempt limit is reached.

We used three datasets from prior watermarking work [14]: (1) **OpenGen**, which asks the underlying LLM to generate continuations of given text (290 samples), (2) **LFQA**, which asks the LLM to answer a question (265 samples), and (3) **FactScore**, in which the LLM generates a biography (291 samples). All experiments use GPT-4.1 as the underlying LLM.

As established in prior work (e.g., [4], [14]), we report true positive rate (TPR) at a fixed false positive rate (FPR) of 1%. We also report the impact of paraphrasing on TPR.

IV. Results

A. TPR Comparison with the Baselines

Table II compares the performance of our watermarking method with the published PostMark results [14]. To ensure a fair comparison, we consider only the results obtained using GPT-4. Our method achieves a true positive rate (TPR) that is approximately one percentage point lower than PostMark; however, both methods remain close to 100%, indicating that our RST-based approach is a competitive alternative to statistical watermarking techniques. As expected, our method is more robust to paraphrasing. Notably, from the results reported in the PostMark paper (reproduced here on Figure 4) it can be seen that both PostMark and our approach are much less sensitive to paraphrasing than all statistical watermarking baselines, with the exception of the Unigram model.

B. Ablation Tests

We conducted a series of ablation studies on our system using 25 prompts from each dataset, namely OpenGen, FactScore, and LFQA, for a total of 75 test samples. All experiments were performed on the same set of 75 samples using GPT-4.1. The results are reported in Table III. Our findings show that removing the relation definitions has the largest negative effect on the false positive rate (FPR), increasing it from 1% to 5.33%. This may be due to a rise in hallucinations caused by the reduced contextual information available to the model. Despite this change, the system still performs well in detecting watermarked text, achieving a true positive rate (TPR) of 96%. However, the increase in FPR suggests that, without the additional context, the model is more likely to overestimate the score (Figure 5a). The largest reduction in TPR occurs when the penalties and the target score for the embedder are removed, lowering the TPR to 86%. This corresponds to a drop of 12.6% across all three test sets, as shown in Table III.

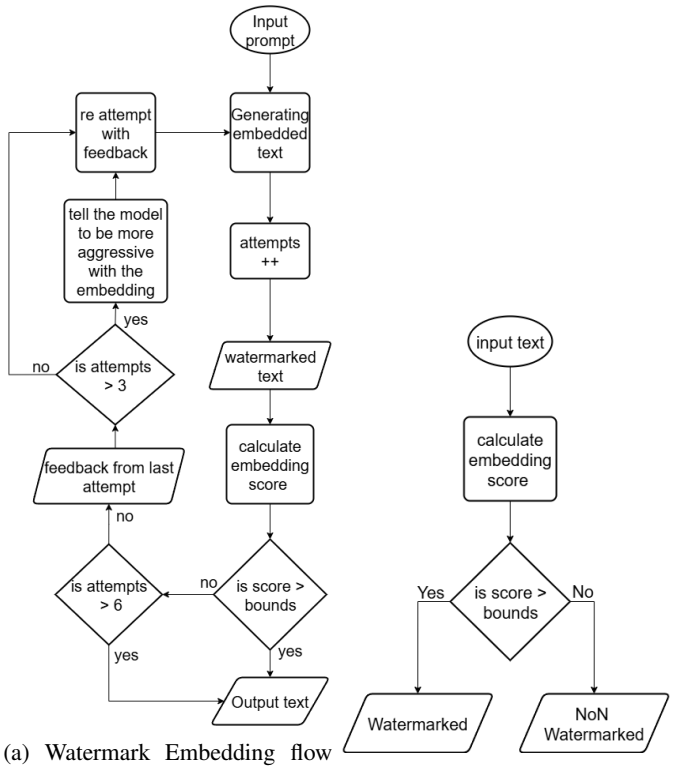
C. Impact of Watermark on Text Quality

Using RST relations as watermarks means that the model is not constrained to insert specific characters or words to encode the watermark. Instead, the watermark operates at the level of text structure and discourse flow. This structure can, in principle, be customized for different use cases by assigning different scores and positional modifiers, although that setting

[1]https://github.com/charlieMcM/Discourse-Level-Watermarking-using-Rhetorical-Set-Theory

Relation	Nucleus (N)	Satellite (S)
Explanation	The road is wet.	It rained overnight.
Evidence	The cake is done.	A toothpick comes out clean.
Contrast	The book is short.	Yet it covers every key plot point.
Background	She packed a jacket.	The forecast predicted cold weather.
Condition	The picnic will happen.	If the rain stops.

TABLE I: Examples of RST relations.



(a) Watermark Embedding flow chart

(b) Extraction flow chart

Fig. 1: Watermark detection and embedding flow charts

TABLE II: True Positive Rate (TPR) at 1% FPR across datasets (before paraphrasing / after paraphrasing) using GPT-4.1

Dataset	RST watermarking	PostMark @12
lfqa	96.6% / 66.8%	99.4% / 65%
openGen	99.0% / 76.9%	97.0% / 59.4%
factScore	100% / 88.0%	-
combined	98.6% / 77.5%	98.2% / 62.2%

is not explored here. As a result, the generated watermarked text remains similar to the original not only in meaning but also in length, as shown in Table 2.

Figure 3 presents two closely related texts: the first is watermarked, and the second is a paraphrased version of it. Both texts preserve the same overall meaning and broad discourse structure. Both are identified as having *contrast* and *interpretation* as their central relations, but in the paraphrased version, *restatement* is placed as a *satellite* over a *nucleus*, which reduces its influence. The paraphrased text therefore preserves the same rhetorical intent, but it also flattens some explicit discourse markers and reduces relation density. A detection model tailored to rhetorical structure would likely identify this more effectively than a generic model such as GPT, especially if the scoring scheme and selected relations were further refined.

V. Conclusions, Limitations and Future Work

A. Conclusions

This paper introduced a discourse-level, prompt-based watermarking approach that uses Rhetorical Structure Theory (RST) to embed attribution signals through rhetorical relations rather than token distributions. The core theoretical contribution is the observation that many meaning-preserving rewrites preserve discourse intent and organisation more reliably than surface form, and that these discourse relations can therefore act as a semantically grounded carrier for watermark information. Empirically, we show that this design can improve

	OpenGen	LFQA	Factscore
watermarked	US 2's transformation from a patchwork of trails and named routes into a unified federal highway reflects not just technological advancement, but also the changing ambitions of the region. While the scenic allure of Lake Michigan and the enduring legacy of historic trails provide a sense of continuity, the development of paved trunklines marked a sharp shift in accessibility and economic possibility for Michigan's Upper Peninsula. The adoption of the U.S. Highway System in 1926 established US 2 as a modern arterial road, binding together remote towns and fostering new travel patterns. Where previously the journey from Iron Mountain to St. Ignace required navigating local trails and often impassable roads, US 2 offered a reliable means for commerce and tourism to flourish. The rise of automobile travel in the 1920s further accentuated this shift: motorists sought direct, well-marked routes, fueling demand for improved infrastructure and services along the corridor. Yet, the replacement of evocative names like Cloverland Trail with numerical designations also signaled a broader trend—practicality superseding regional romanticism. Although some lamented the fading of distinctive local identities, the streamlined highway system enabled unprecedented mobility. If the old trails embodied individual exploration, US 2's creation realized the collective aim of linking communities and supporting economic growth, demonstrating how transportation policy can both reflect and redefine a region's character.	When exploring the relationship between air pressure changes and temperature, it is crucial to recognize that temperature directly influences the movement of air molecules. As temperature increases, air molecules gain kinetic energy, move more rapidly, and expand the volume they occupy. Consequently, this expansion leads to a decrease in air density, and if the air is not confined, the air pressure drops; this outcome occurs because the same number of molecules exert force over a larger area. Conversely, when temperature decreases, air molecules lose energy, move more slowly, and contract, thus increasing air density and resulting in higher air pressure—a clear antithesis that highlights the opposing effects of temperature increases and decreases.Such dynamics are fundamental to atmospheric processes: non-volitional cause reveals that warm air rising at the Earth's surface (caused by solar heating) leads to lower air pressure regions, while cooler air descending causes higher pressure zones. If these pressure differences did not form, wind patterns and weather systems as we know them would not exist—an otherwise relation, since preserving a constant temperature would prevent the essential redistribution of air masses.In practical scenarios, meteorologists rely on evidence relating pressure and temperature to forecast weather changes. For instance, a rapid drop in air pressure is often a precursor to storms, justified by the close link between temperature-induced pressure changes and unstable weather conditions. Interpreting these fluctuations within the broader framework of atmospheric science underscores the purpose of studying air pressure and temperature: to predict and understand shifting weather, emphasizing their interdependency and impact on our environment.	Bobby Fischer's life is a fascinating study in both brilliance and paradox. While he is celebrated as one of the greatest chess players in history, his journey was also marked by extraordinary personal struggles. What distinguishes Fischer is not only his unparalleled achievements on the chessboard but also the contradictions that shaped his legacy.If Fischer had not devoted himself so relentlessly to chess, his genius might have remained undiscovered. Yet, his obsession with the game provided the means for him to transform the world of competitive chess. The pursuit of mastery became his singular purpose, driving him from a young prodigy to the 11th World Chess Champion.Fischer's rise began in childhood; after learning chess at age six, he quickly outstripped local competition. As he progressed through the ranks, his victories at the U.S. Championship—beginning at age 14—proved that his talent was both authentic and prodigious. His unwavering focus, combined with an innovative, aggressive style, changed how players approached the game. The chess community recognized Fischer not only for his technical prowess but also for his psychological intensity.However, unlike many champions who thrive on collaboration and community, Fischer often isolated himself. This tendency both enabled him to hone his skills in solitude and contributed to the mythos of the "lone genius." Moreover, millions followed his matches, particularly during his 1972 showdown with Boris Spassky, which became a symbolic battle of East versus West. In this match, Fischer's victory ended decades of Soviet dominance; this result was remarkable not only for its impact on international chess but for its broader cultural implications.Nevertheless, Fischer's later life contrasted sharply with his earlier triumphs. As his erratic behavior intensified and his interactions with the chess world diminished, Fischer drifted from the spotlight. If he had chosen a more conventional public life, he might have further contributed to chess as a teacher or ambassador. Otherwise, his withdrawal insulated him from both the pressures and the accolades associated with fame.Beyond chess, Fischer's personality was interpreted in many ways. Some regarded his anti-establishment attitude and controversial views as evidence of a troubled mind. Others believed his genius came at the cost of personal peace. Fischer's legacy, therefore, exists in a framework that places his achievements against the backdrop of his tumultuous personal journey. Through his brilliance and his contradictions, Bobby Fischer remains a symbol of both the heights and the costs of human potential.
Non Watermarked	By the mid-1920s, as motor travel grew in popularity, the Cloverland Trail became part of a broader system of named auto trails crisscrossing Michigan and the nation. Among the most prominent was the Theodore Roosevelt International Highway, extending from Portland, Maine, to Portland, Oregon, and using the corridor of present-day US 2 across Michigan's Upper Peninsula. As the federal government introduced the numbered U.S. Highway System in 1926, these local and regional trail names faded into history, supplanted by more standardized numbering: the eastern segment of the Theodore Roosevelt route in the UP officially became US 2. Road improvements followed the new federal designation, with sections once impassable in spring or rough with corduroy roads gradually upgraded to gravel or paved surfaces by the 1930s.Communities along the route adapted to the changes brought by increased automobile tourism: former lumber towns sought to reinvent themselves as service centers or tourist destinations. Motels, cabins, diners, and gas stations sprang up along the emerging highway, catering to motorists heading to summer resorts or to the Straits of Mackinac. The designation of US 2 also fostered a sense of connection to the broader American highway network, which helped boost the UP's identity as part of a nation on the move. Decades later, while bypasses and realignments have subtly reshaped its course, US 2 remains both a vital corridor for local communities and a scenic gateway for travelers exploring Michigan's north woods and Lake Michigan's dramatic shoreline.	Air pressure and temperature are closely linked in the Earth's atmosphere. Air pressure is the force exerted by the weight of air above a particular point, while temperature refers to how hot or cold the air is. When the temperature increases, air molecules gain energy and move faster, causing the air to expand. As air expands, it becomes less dense and rises, which leads to a decrease in air pressure at the surface. Conversely, when the temperature drops, air molecules lose energy, move closer together, and the air becomes denser. This denser, cooler air sinks toward the ground, increasing the surface air pressure.These changes are fundamental to weather patterns. For example, warm regions, such as the tropics, often have low air pressure because of the rising, heated air, while cooler regions, like the poles, feature higher air pressure due to the sinking cold air. This difference in air pressure causes wind, as air moves from high-pressure (cold) areas to low-pressure (warm) areas, balancing energy across the planet.Furthermore, rapid temperature changes can cause fast shifts in air pressure, sometimes leading to strong winds, storms, or even weather fronts. Understanding the relationship between temperature and air pressure is essential in meteorology because it helps explain everything from daily breezes to the formation of large-scale weather systems like cyclones and anticyclones.	**Bobby Fischer** (Robert James Fischer) was an American chess grandmaster and the eleventh World Chess Champion, widely considered one of the greatest and most influential chess players of all time. ### Early Life- **Born**: March 9, 1943, Chicago, Illinois, USA - **Mother**: Regina Wender Fischer, a teacher and nurse - **Father**: Officially listed as German biophysicist Hans-Gerhardt Fischer, but evidence suggests his biological father might have been Paul Nemenyi, a Hungarian physicist.Fischer grew up in Brooklyn, New York, with his mother and elder sister Joan. He learned the rules of chess at age six and quickly developed a prodigious talent.### Rise to FameFischer's precocity was evident as he won the U.S. Junior Chess Championship at age 13. At 14, he became the youngest U.S. Chess Champion and represented the United States at the Chess Olympiad.### Achievements- **Grandmaster Title:** In 1958, Fischer became the youngest grandmaster in history (age 15 years, 6 months), a record at the time.- **Eight-time U.S. Champion:** He won every U.S. Championship he entered, often by wide margins.- **1964 U.S. Championship:** Finished with a perfect score (11–0).- **1970 Interzonal and Candidates Matches:** Fischer scored unprecedented victories, winning 20 consecutive games at the highest level.### World Chess Champion- **1972 World Chess Championship:** Fischer defeated Boris Spassky of the Soviet Union in Reykjavik, Iceland. This match was a symbolic Cold War event and made Fischer a worldwide celebrity. He was the first American-born player to become World Champion.### Later Life and ControversiesFischer declined to defend his title in 1975 after FIDE did not grant his match conditions — Anatoly Karpov became World Champion by default. Fischer became reclusive and disappeared from public view for years.In 1992, Fischer played a rematch against Spassky in Yugoslavia, violating U.S. sanctions. This led to an arrest warrant in the U.S. and further isolation.Fischer spent his later years in Hungary, the Philippines, and Japan, making occasional anti-American and anti-Semitic statements. He was detained in Japan in 2004 but eventually granted asylum in Iceland.### Death- **Died:** January 17, 2008, Reykjavik, Iceland, from kidney failure.### LegacyBobby Fischer revolutionized chess with his deep understandings of theory, demands for better playing conditions, and independent approach. His book, *My 60 Memorable Games*, remains a classic. Despite his personal controversies, Fischer's impact on chess and its popularization in the United States remains unparalleled.---**References:** - [Bobby Fischer – Wikipedia] (https://en.wikipedia.org/wiki/Bobby_Fischer) - *Bobby Fischer Goes to War* by David Edmonds and John Eidinow - *Endgame: Bobby Fischer's Remarkable Rise and Fall* by Frank Brady

Fig. 2: Pairs of watermarked and non-watermarked texts from all 3 datasets.

watermarked text	paraphrased text	extraction results
The relief of Kimberley marked a pivotal moment, as Australian and British forces coordinated a swift advance to break Boer encirclement. While the Boers strove to maintain their advantage elsewhere, their defensive lines around Kimberley were ultimately breached, compelling a rapid withdrawal. Although the operation achieved its purpose, it highlighted the evolving nature of the conflict: open-field engagements were giving way to guerrilla tactics as Boer commanders responded to the superior resources and numbers of their adversaries. If the Boers had managed to retain control of Kimberley, the war's duration might have been prolonged and British supply lines threatened, but the successful relief shifted momentum decisively in the British favour.With victory at Kimberley, Australian mounted forces—now recognized for their adaptability and horsemanship—were increasingly tasked with reconnaissance and pursuit operations. However, these successes stood in stark contrast to the escalating hardships faced by soldiers and civilians alike. For instance, as the British extended their campaign into the Orange Free State, they resorted to the controversial "scorched earth" policy and established concentration camps for Boer families, intending to undermine the support base of the commandos. This deliberate targeting of non-combatants provoked international condemnation and forced a reassessment of the war's moral dimensions, even while the military objectives—namely, the eventual surrender of the Boer Republics—remained unchanged.The adaptation of both military strategy and public sentiment underscored the deep contrasts within the conflict. The Australians, who initially embraced the adventure, soon confronted the grim realities of modern warfare. Nevertheless, their continued engagement played a significant role in consolidating imperial control in South Africa, while setting a precedent for future Australian expeditionary contributions around the empire.	The liberation of Kimberley was a turning point, with British and Australian troops working together to quickly break through the Boer siege.Although the Boers tried to hold their ground in other areas, their fortifications encircling Kimberley were eventually overcome, forcing them to retreat swiftly.While the mission was successful, it underscored a shift in the war, as traditional battles were replaced by guerrilla warfare in response to the enemy's greater manpower and supplies.Had the Boers succeeded in holding onto Kimberley, the conflict could have lasted longer and put British supply routes at risk; however, the victory at Kimberley gave the British a clear advantage and changed the course of the war.Following the triumph at Kimberley, Australian mounted troops—renowned for their flexibility and riding skills—were more frequently assigned to scouting and chasing the enemy.Nevertheless, these victories sharply contrasted with the increasing difficulties experienced by both soldiers and civilians.For example, when the British advanced into the Orange Free State, they implemented the contentious "scorched earth" strategy and set up concentration camps for Boer families, aiming to weaken the commandos' backing.The intentional mistreatment of civilians drew widespread criticism from the global community and compelled a reevaluation of the ethical aspects of the conflict, despite the fact that the primary military goal—the capitulation of the Boer Republics—stayed the same.Changes in tactics and shifts in public opinion highlighted the profound divisions present in the war.Although the Australians were at first enthusiastic about the adventure, they quickly faced the harsh truths of contemporary warfare.Even so, their ongoing involvement was instrumental in strengthening imperial authority in South Africa and established a model for later Australian participation in overseas imperial campaigns.	<Watermarked> with score 48 (with explanation) [Explanation: The passage claims a score of 48 per 100 words, which exceeds the required threshold (N > 45) for a clear watermark. The explanation details several high-weight Rhetorical Structure Theory (RST) relations—CONTRAST, CONDITION, NON-VOLITIONAL CAUSE, CONCESSION, ELABORATION, EVALUATION, SEQUENCE, and SUMMARY. Especially notable is the frequent and varied use of CONTRAST near the root and key thematic pivots, as well as deep narrative embedding of other relations (mid/upper-level for CONCESSION, EVALUATION, and SEQUENCE). There are at least four distinct, non-singleton relation families, with strong hierarchical and positional structuring, supporting the robustness of the watermark signal. No penalties apply, and the score is normalized correctly. The explanation also proposes advanced improvements but acknowledges the already "strong relational structuring." All these factors indicate a clear, strong watermark.]

Fig. 3: Example of watermarked text before and after paraphrasing along with the detection report.

Metric → Model ↓	Dataset ↓	Avg Entropy ↓	TPR at 1% FPR (Before Paraphrasing / After Paraphrasing)								
			PostMark@12	Blackbox	KGW	Unigram	EXP	EXP-Edit	SIR	SemStamp	k-SemStamp
Llama-3-8B	OpenGen	3.6	99.7 / 63.5	81.2 / 2.2	100 / 74.8	99.8 / **93.4**	99.8 / 36.6	97.3 / 73.3	-	-	-
	LFQA	3.5	97.8 / 72.5	82.8 / 1.6	99.8 / 25.6	99.8 / **79.6**	99.8 / 12.4	83 / 41	-	-	-
Llama-3-8B-Inst	OpenGen	1.6	99.4 / **46.4**	91.8 / 1	98.2 / 21.6	99.6 / 41.4	99.6 / 4.8	47.8 / 2.2	-	-	-
	LFQA	1.3	96 / **65.7**	86.2 / 3	85.8 / 19	98.6 / 31.8	98.4 / 0.6	21.1 / 0.6	-	-	-
Mistral-7B-Inst	OpenGen	1.4	99.2 / **69.2**	98.4 / 0.4	100 / 16	99.8 / 56	99.4 / 5	33 / 1.5	-	-	-
	LFQA	1.1	99.6 / **56.4**	89.8 / 0.4	99.4 / 23.6	97.2 / 41.2	97.4 / 0.8	20.1 / 2.1	-	-	-
GPT-4	OpenGen	-	99.4 / **59.4**	99.4 / 1.4	-	-	-	-	-	-	-
	LFQA	-	99.4 / **65**	99.2 / 0.4	-	-	-	-	-	-	-
OPT-1.3B	RealNews	3.6	98.2 / 67.2	1.2 / 0	99.2 / 40.8	98.8 / 77.2	99.4 / **80.7**	69.6 / 46.7	99.4 / 24.7	93.9 / 33.9	98.1 / 55.5

Fig. 4: Comparison between PostMark and lexical watermarking approaches, reproduced from [14].

TABLE III: Ablation testing result over all 3 datasets using GPT-4.1

Test	TPR	FPR
Relation Removal	96%	**5.33%**
No target	92%	4%
No Penalties	92%	0%
No target and Penalties	**86%**	0%

robustness to paraphrasing while maintaining output quality, and can do so without requiring access to model internals.

Our method is motivated by the detectability–robustness–quality trade-off that has limited prior watermarking. Token-level schemes can be highly detectable but are often brittle to paraphrase, whereas aggressive robustness can harm fluency or meaning. By shifting the carrier to discourse structure, we target robustness to paraphrasing without forcing unnatural lexical constraints. We also retain interpretability, as the watermark is expressed in terms of explicit rhetorical relations that can be inspected and audited.

B. Limitations

First, our evaluation relies primarily on GPT-4.1 for generation and scoring, and we do not provide cross-model validation. This limits claims about portability to smaller or differently trained LLMs (for example, open-weight models) and to systems with weaker instruction following. The approach is prompt-based and therefore in principle model-agnostic, but practical performance will depend on whether a given model can (i) follow relation-guidance prompts and (ii) express the intended discourse patterns with sufficient consistency.

Second, our adversarial testing is limited to single-pass paraphrasing. We do not evaluate stronger, adaptive attacks such as multi-round paraphrasing, summarisation followed by rephrasing, cross-lingual back-translation, or mixed transformations that explicitly target discourse structure. These attacks may reduce detectability further and could reveal failure modes that do not arise under generic paraphrase.

Third, scalability remains untested. Our experiments focus on relatively short texts (up to 250 words) and a small set of genres. It is unclear how relation density and the watermark objective scale with longer documents, and whether the same prompting strategy remains stable for genres that deviate from expository prose (for example, poetry, code, or highly technical writing). Longer texts also introduce more degrees of freedom for both the inserter and an attacker, which may affect detectability and robustness.

Fourth, while we use a weighted relation set in practice, we do not provide a full analysis explaining why particular relations are especially suitable watermark carriers, nor do we isolate how relation selection drives robustness. In particular, relations with clear discourse functions (for example, contrast or cause-like relations) may provide stronger constraints on discourse organisation than looser relations, but our current study does not systematically test this hypothesis.

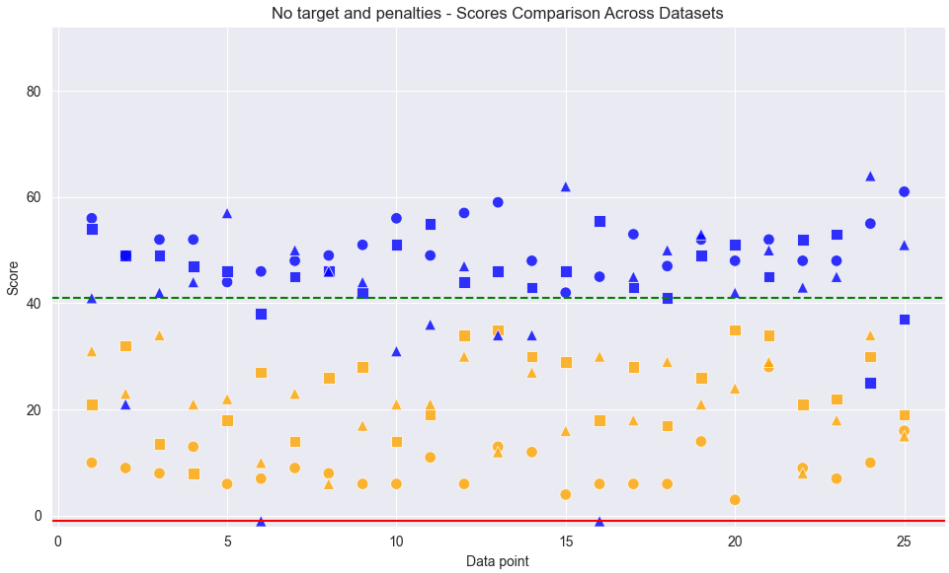
Finally, we do not report quantitative RST diagnostics that directly measure whether the generated text realises the intended RST pattern. Because our pipeline is prompt-based, the model may produce text that appears coherent while only partially satisfying the designed relation plan, or it may implicitly "hallucinate" relations that are not supported by the text in a strict discourse-analytic sense. We therefore cannot fully characterise (i) relation density realised in the output, (ii) consistency of relation placement across generations, or (iii) accuracy of any automated relation labelling used within the pipeline.

C. Future Work

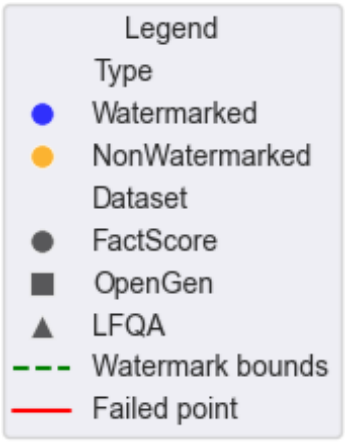
Several directions follow naturally from these limitations and are actionable. (1) Cross-model validation: evaluate the method across closed and open models (including smaller instruction-tuned LLMs) to understand how model capability affects embedding and detection, and to identify prompt



(a) Relation Removal



(b) No target and Penalties



(c) Legend

Fig. 5: Ablation results with shared legend.

designs that transfer. (2) Stronger adversarial evaluation: test multi-round paraphrasing, summarisation-plus-rewrite, and cross-lingual transformations, and include adaptive attackers that explicitly aim to alter discourse relations while preserving meaning. (3) Scalability studies: extend evaluation to longer documents and diverse genres, and investigate how to normalise watermark strength as length increases.

A particularly promising direction is to replace or augment prompt-only discourse control with a dedicated discourse component. This could include fine-tuning a discourse parser or relation classifier on RST-style corpora, then using it as an external constraint or reward model to guide generation toward desired relation profiles. Such a component would also enable quantitative RST analysis, for example reporting realised relation densities, parser confidence, and agreement between intended and inferred relations. Finally, cross-language validation using multilingual discourse resources could clarify whether discourse-level carriers remain stable under translation and whether multilingual attackers can reliably remove the watermark.

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VI. Appendix

The following prompts were used for each dataset. Prompts and dataset were taken from [14]:

- **Opengen:**
  `prompt = "Generate a continuation for the following text in 200-250 words, only include the continuation in your response.\n\n{ex['prefix']}"`
- **LFQA (Large Format Question Answers):**
  `prompt = "Answer the following question in 200-250 words.\n\n{ex['prefix']}"`
- **Factscore:**
  `prompt = "Biography of {ex['title']}:"`
- **paraphrasing:**
  `prompt = "Given some previous context and a sentence following that context, paraphrase the current sentence. Only return the paraphrased sentence in your response.`
  `Previous context: {pText}`
  `Current sentence to paraphrase: {sentence}.`
  `Your paraphrase of the current sentence:"`